SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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Register Number	**192373066**

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I A. Lakshmi(192373066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Lakshmi

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Answers:

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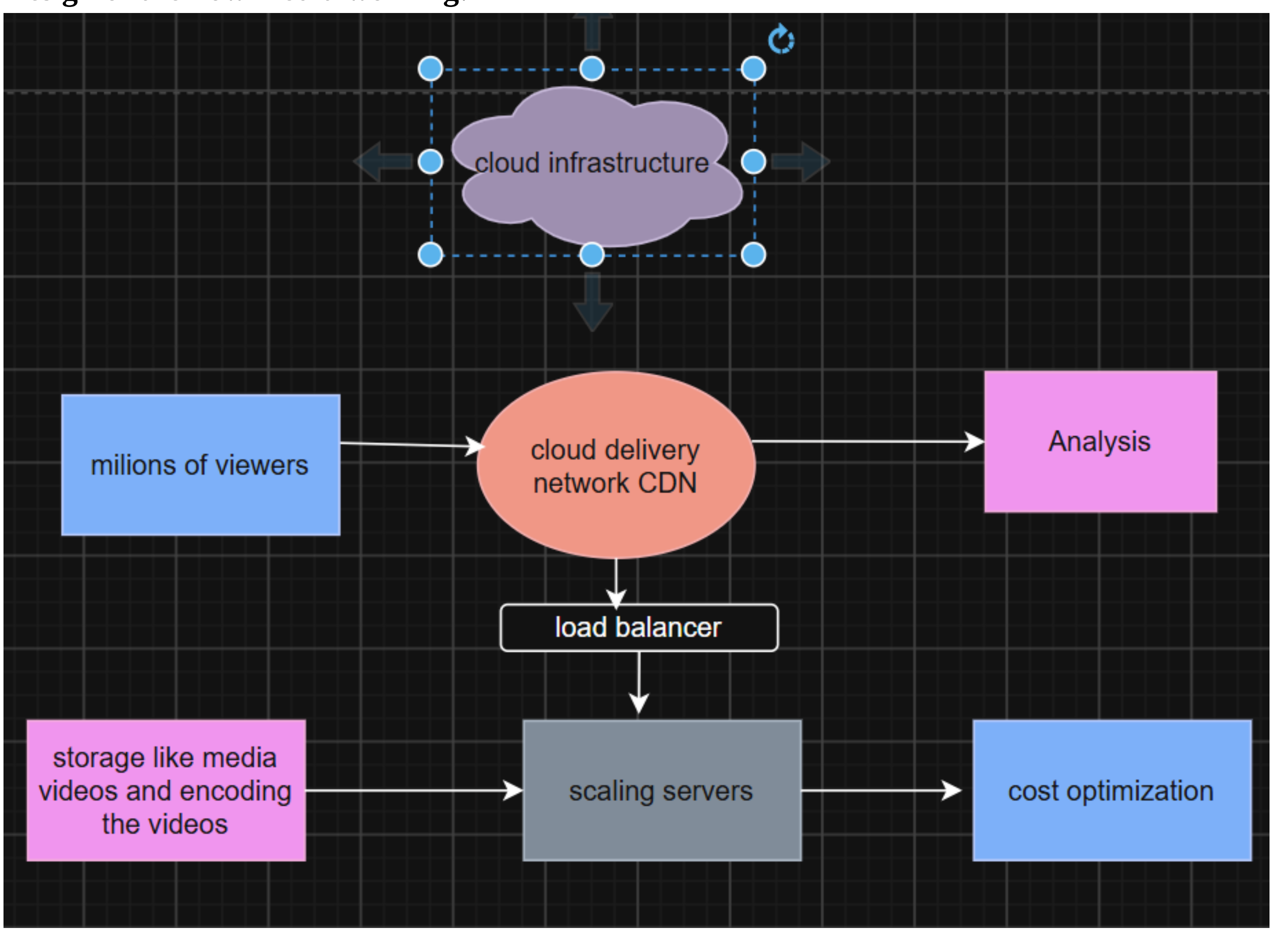
1.Media Company – Cloud Video Streaming

Imagine millions of viewers tuning in for a live concert.

- **Architecture: Using** CDN (content Delivery Network) as a "front row", catching all media information like it is close to the users wish. Behind it, a fleet of a streaming servers sits in auto -scaling groups, ready to expand like an according when traffic spikes
- **Load Balancing:** Think of it as the bouncer at the club which is going to directing users to the least crowded door so nobody waits too long.
- **Auto-Scaling:** When a new blockbuster drops, servers spin up automatically, then shrink back down when the hype fades.
- **Cost Optimization:** Use spot instances for non-critical workloads, popular and unpopular, and predictive scaling so you don't pay for all non-used servers.

Design of the how media working:



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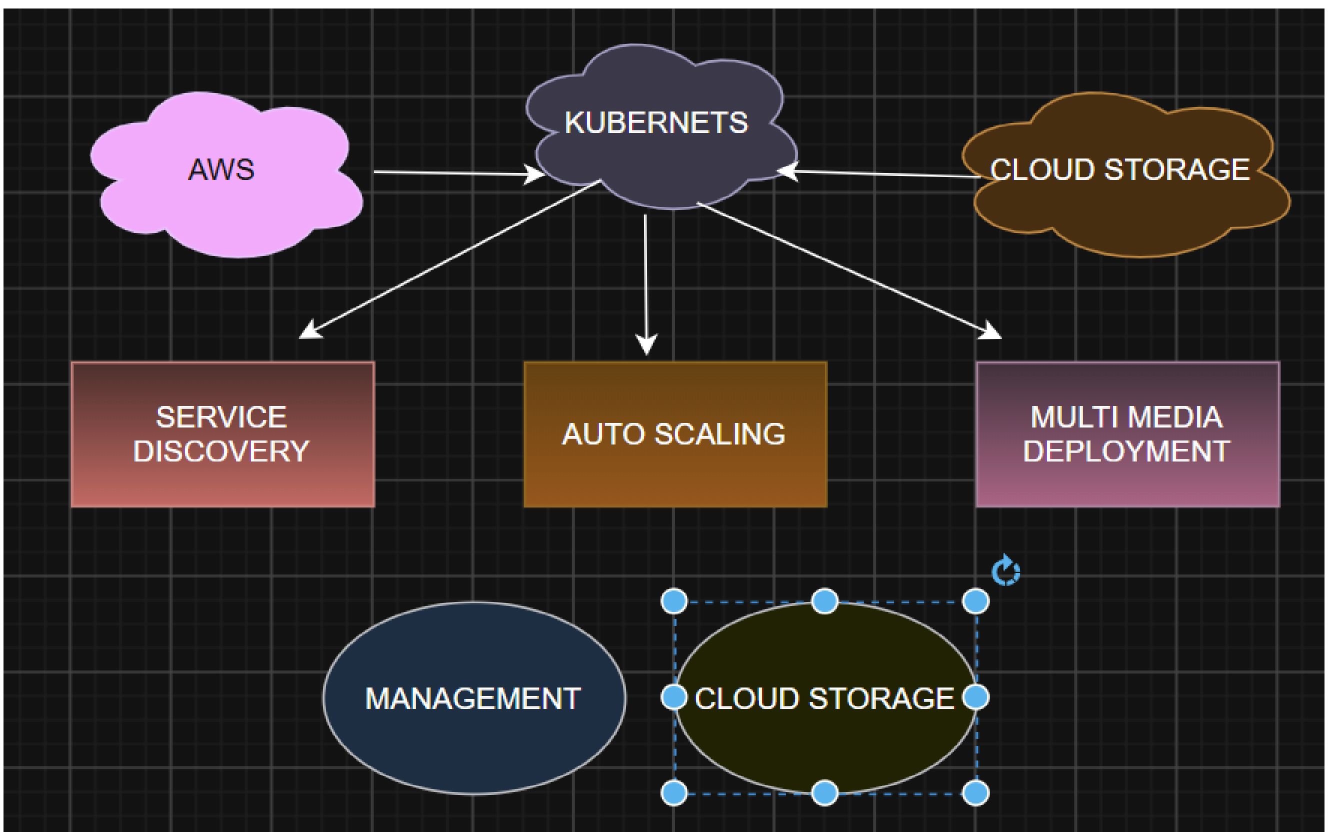
Picture their apps as shipping containers on cargo ships sailing across AWS, Azure, and GCP.

- **Solution:** Kubernetes acts as the harbor master, ensuring containers are deployed consistently no matter the cloud.
- **Service Discovery & Scaling:** With Kubernetes services and ingress controllers, apps find each other by their features . Horizontal Pod Autoscaling keeps things unbreakable like the elastic bond
- **Benefits:** No vendor lock-in, resilience if one provider stumbles.
- **Risks:** Complexity skyrockets—like juggling three phones at once. Networking costs and management overhead can bite if not planned carefully.

Design the deployement:



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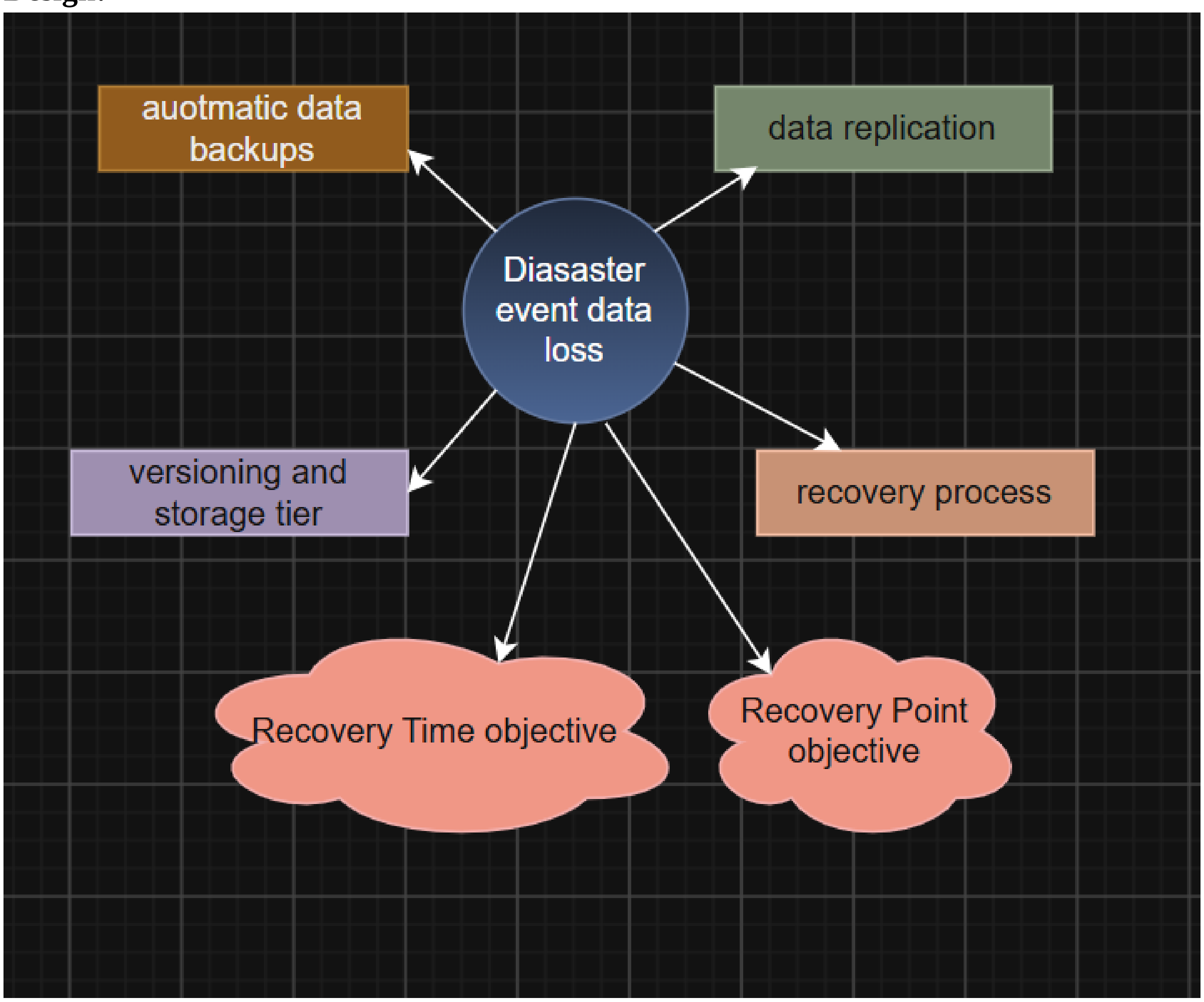
Accidental deletion feels like losing your homework file the night before submission.

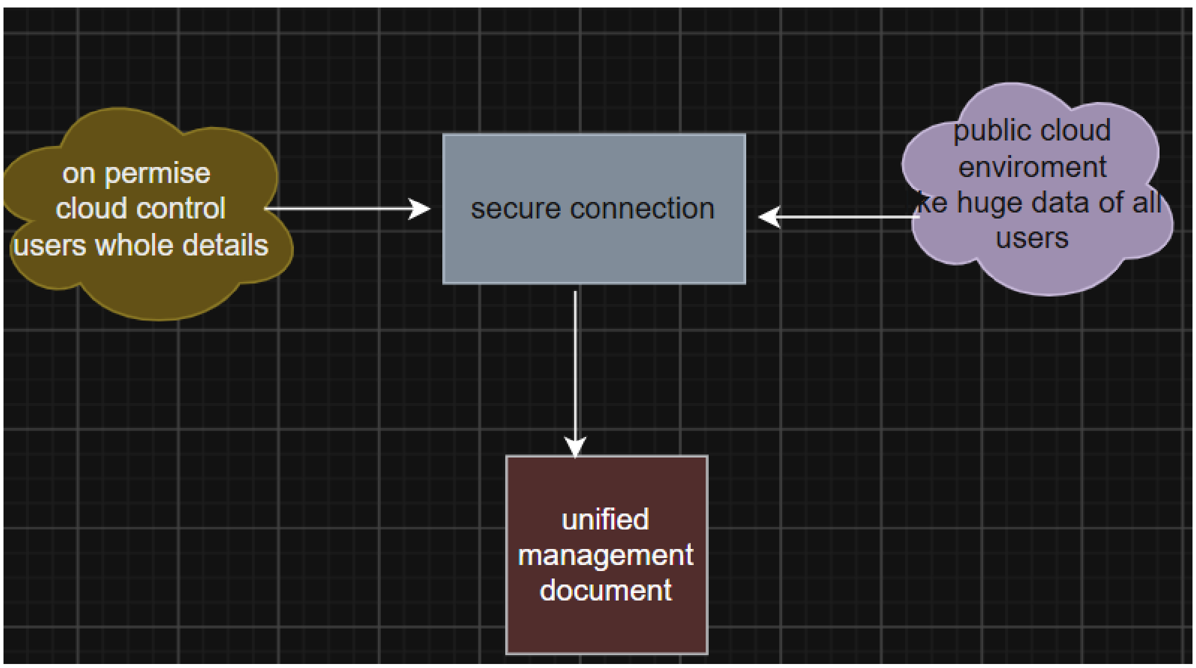
- **Solution:** Enable versioning in object storage (like S3), replicate data across regions, and schedule automated backups.
- **RTO & RPO:**
  - ➢ RTO (Recovery Time Objective): Snapshots and replicas mean you can restore in minutes.
  - ➢ RPO (Recovery Point Objective): Frequent backups ensure you only lose seconds or minutes of data, not hours.
- **Business Continuity:** Test recovery drills regularly—like fire drills for data. Keep failover environments warm so students never miss a class.

Design:



4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Sensitive data stays locked in the on-prem vault, while non-sensitive workloads enjoy the cloud's playground.

- **Architecture:** Use secure VPNs or dedicated connections (like Azure ExpressRoute) to bridge environments.
- **Identity Management:** Federated identity (Azure AD/Okta) ensures users log in once and access both worlds securely.
- **Encryption:** Encrypt data at rest and in transit—think of it as armored trucks moving between vaults.
- **Challenges:** Governance complexity, latency between environments, and ensuring compliance across two very different worlds.
- 

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

SaaS Provider – Multi-Region High Availability

Imagine your app as a global concert tour—it must play on even if one city's stage collapses.

- **Strategy:** Deploy in multiple regions with active-active architecture. If one region fails, traffic reroutes instantly.
- **Data Replication:** Use database replication (Aurora Global Database, Cosmos DB) so data is always in sync.
- **Failover:** DNS-based failover (Route 53, Traffic Manager) ensures users are redirected seamlessly.
- **Monitoring & Alerts:** Think of it as backstage crew—tools like Prometheus, Grafana, and CloudWatch constantly watch for issues and send alerts before the crowd notices.